Solutions to JEE Advanced -9 |JEE 2022 | Paper-1

PHYSICS

1.(D) $Ri(t) = \varepsilon_{ind}(t) - V_C,$

$\Rightarrow Ri(t) = BLv(t) - \dfrac{Q(t)}{C}.$

$BLv_{\min} = \dfrac{Q_{\max}}{C}.$ ..(1)

$m\Delta v = F\Delta t.$

$m\Delta v_n = -BLi\Delta t.$

$i\Delta t = \Delta Q,$; $m\Delta v_n = BL\Delta Q_n.$

$m\sum_n \Delta v_n = -BL\sum_n \Delta Q_n,$

$m(V_{\min} - V_0) = -BLQ_{\max}.$ …(2)

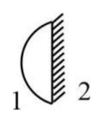
Expressing $Q_{\max}$ from equation (1) $Q_{\max} = CBLv_{\min},$

And writing it into equation (2), $mV_{\min} - mV_0 = -BLCBLv_{\min} = -B^2L^2Cv_{\min},$

We get that the final constant speed of the rod is: $V_{\min} = \dfrac{mV_0}{m + (BL)^2 C}.$

2.(C) Using the equation, $\dfrac{1}{F} = \dfrac{2(\mu_2/\mu_1)}{R_2} - 2\dfrac{(\mu_2/\mu_1 - 1)}{R_1}$

We get $\dfrac{1}{-10} = \dfrac{2(1.5)}{\infty} - \dfrac{2(1.5-1)}{R_1}$, $\left(\text{use } F = \dfrac{R}{2}\right)$



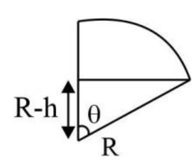
Solving, we get $R_1 = 10cm.$

3.(B) $1\,SD = \dfrac{10}{200}cm = 0.050cm;$ $1VD = \dfrac{1.2}{25}cm = 0.048cm.$

LC of Instrument = 0.050 – 0.048 = 0.002 cm

4.(C) Condition for leaving contact

$mg\left(\dfrac{R-h}{R}\right) = \dfrac{mv^2}{R}$ …(i)



From work-energy theorem $2mgh = \dfrac{1}{2}mv^2$

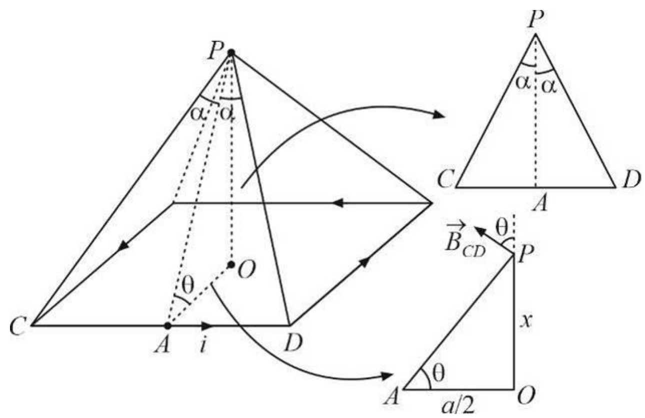
$mv^2 = 4mgh$ …(ii)

From (i) and (ii) we get $h = \dfrac{R}{5} = 40cm$

5.(A) Let us consider magnetic field at point '*P*' due to wire *CD*.

$$AP = \sqrt{x^2 + \frac{a^2}{4}} = \frac{\sqrt{4x^2 + a^2}}{2}$$

$$CP = \sqrt{AC^2 + AP^2} = \sqrt{x^2 + \frac{a^2}{2}} = \frac{\sqrt{4x^2 + 2a^2}}{2}$$



$$\sin\alpha = \frac{AC}{CP} = \frac{a}{\sqrt{4x^2 + 2a^2}}$$

Now, $$B = \frac{\mu_0 i}{4\pi(AP)} 2\sin\alpha = \frac{\mu_0 ia}{\pi\sqrt{4x^2 + a^2}\sqrt{4x^2 + 2a^2}}$$

When the fields of all the four sides are considered, the horizontal components add to zero. So, the total field is given by

$$B_R = 4B\cos\theta = \frac{4\mu_0 ia^2}{\pi(4x^2 + a^2)\sqrt{4x^2 + 2a^2}}$$

$$B_R = \frac{4\mu_0 ia^2}{\pi(4x^2 + a^2)\sqrt{4x^2 + 2a^2}}$$

$$= \frac{4\mu_0 \times 1 \times 2}{\pi(4\times 3 + 2)\sqrt{4\times 3 + 2\times 2}} = \frac{8\mu_0}{\pi \times 14 \times 4} = \frac{\mu_0}{7\pi}$$

6.(A) At this instant, potential on sphere is zero

Let q' be charge induced on sphere. So potential on sphere $= \frac{kq}{x} + \frac{kq'}{R} = 0$

$$q' = -\frac{qR}{x}$$

$$\frac{dq'}{dt} = -\frac{qR}{x^2}\frac{dx}{dt}$$

$$i = \frac{qR}{x^2}V_0$$

7.(A) $t \propto \frac{d}{A}$, $t' \propto \frac{d/2}{2A}$

Dividing, $\frac{t'}{t} = \frac{1}{4}$ $or\ t' = \frac{t}{4} = \frac{4\,\text{min.}}{4} = 1\,\text{min}.$

8.(A) $E = nhv$

$E = \frac{nhc}{\lambda}$ or $n = \frac{E\lambda}{hc}$

Now, $n' = \frac{n}{100} = \frac{39.8 \times 5000 \times 10^{-10}}{100 \times 6.6 \times 10^{-34} \times 3 \times 10^{8}} \approx 10^{18}$

9.(A) $\frac{I_1}{I_2} = \frac{4}{1}$

$$\frac{I_{max.}}{I_{min.}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = \frac{(2+1)^2}{(2-1)^2} = 9$$

10.(B) The resistance between the electrodes is $R = \int_a^b \frac{\rho dz}{4\pi z^2} . = \frac{\rho(b-a)}{4\pi ab}$

The voltage across the capacitor is $v = v_0 e^{-t/RC}$

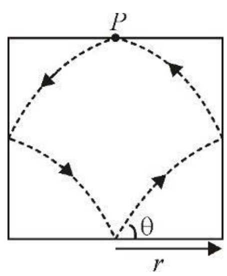
Here $v = \frac{v_0}{n}$ when $t = \tau$; $\frac{v_0}{n} = v_0 e^{-\tau/RC}$

$\boxed{\tau = RC \log_e n}$

$\tau = \frac{\rho(b-a)C}{4\pi ab} \log_e n$; $\boxed{\rho = \frac{4\pi ab\tau}{(b-a)c \log_e n}}$

11.(BD) Range $= 4r = u \cos\theta \times T$ (Speed in horizontal direction remains unchanged)

$u = \frac{4r}{T \cos\theta}$ (as all collision are elastic)



Due to collision at point P with the ceiling velocity in the y direction is reversed, thereby decreasing the time of flight if compared to the scenario when there was no ceiling.

Range (no ceiling) $= \frac{u^2 \sin 2\theta}{g}$

Range (no ceiling) > Range (with ceiling)

$\frac{u^2 \sin 2\theta}{g} > 4r$; $u > 2\sqrt{\frac{gr}{\sin 2\theta}}$

12.(ABC)

From the given expression for y : amplitude A = 0.02 m

Angular frequency $\omega = 50\pi\ rad/s$

and wave number $k = 10\pi\ m^{-1}$

Now wave speed $v = \frac{\omega}{k} = \frac{50\pi}{10\pi} = 5m/s$

Displacement node occurs at $10\pi x = \frac{\pi}{2}, \frac{3\pi}{2} etc.$

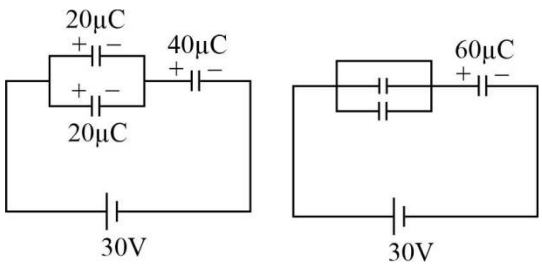
or $x = \frac{1}{20}, \frac{3}{20}$ or $x = 0.05m$ and $0.15m$

Displacement antinode occurs at $10\pi x = 0, \pi, 2\pi, 3\pi \; etc.$

or $x = 0, 0.1\,m,$ 0.2 m and 0.3 m

Wavelength $\lambda = 2$ (distance between two consecutive nodes or antinodes) $= 2(0.1) = 0.2m$

13.(ACD) The charges stored in different capacitors before and after closing the switch S are



The amount of charge flown through the battery is $q = 20\mu C$

$\therefore$ Energy supplied by the battery is $U = qV = \left(20 \times 10^{-6}\right)(30)\,J$

$U = 0.6mJ$

Energy stored in all the capacitors before closing the switch S is

$$U_i = \frac{1}{2}C_{net}V^2 = \frac{1}{2}\left(\frac{4}{3}\times 10^{-6}\right)(30)^2 = 0.6\,mJ$$

and after closing the switch $U_f = \frac{1}{2}C_{net}V^2 = \frac{1}{2}\left(2\times 10^{-6}\right)(30)^2 = 0.9mJ$

$\therefore$ Heat generated $H = U_f - U_i = 0.3mJ$

and charge flown through the switch is $60\mu C.$

14.(ABCD) Work done during isobaric expansion $= P_A(V_B - V_A)$

Work done during aidabic expansion $= \frac{P_BV_B - P_CV_C}{\gamma - 1} = 2(P_BV_B - P_CV_C)$

Work is equal.

$\Rightarrow \quad P_AV_B - P_AV_A = 2P_BV_B - 2P_CV_C \quad \ldots\ldots\ldots$ (i)

Now, $P_C = \frac{8}{27}P_A = \frac{8}{27}P_B \quad \Rightarrow \quad$ using $P_BV_B^{\gamma} = P_CV_C^{\gamma}$, we get $V_C = \frac{9}{4}V_B$

Putting this, $P_B = P_A$ and $P_C = \frac{8}{27}P_A$ in (i), $P_AV_B - P_AV_A = 2P_AV_B - 2\left(\frac{8}{27}P_A\right)\left(\frac{9}{4}V_B\right)$

$\Rightarrow \quad V_B = 3V_A$

$\Rightarrow \quad$ Work, $W = P_A(3V_A) - P_AV_A = 2P_AV_A$

$$\Delta Q_{A\to B} = nC_p\Delta T = \frac{C_P}{R}(P_BV_B - P_AV_A) = 6P_AV_A$$

Also, $V_C = \frac{27}{4}V_A$

15.(BD) $V_{cm} = A_{cm} \times t$

$\omega = \alpha t$

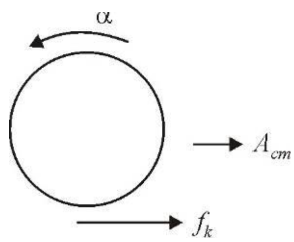
$V_{bottom} = V_{cm} + \omega R$

When pure rolling begins

$V_{bottom} = V_0 = A_{cm}t + R(\alpha t)$

$V_0 = t(A_{cm} + \alpha R)$

Using dynamics …..



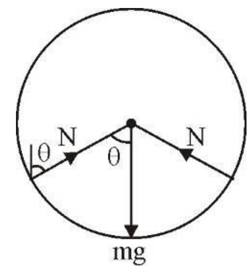
$\mu_k mg = mA_{cm} \Rightarrow A_{cm} = \mu_k g$

$$\mu_k mg \times R = \frac{2}{5}mR^2\alpha \Rightarrow \alpha = \frac{5}{2}\frac{\mu_k g}{R}$$

$$V_0 = t\left(\mu_k g + \frac{5}{2}\mu_k g\right)$$

$$t = \frac{2V_0}{7\mu_k g}; \qquad V_{cm} = \mu_k g \times t = \frac{2V_0}{7}$$

16.(16) Normal reaction exerted by the slab will be radially inwards towards the centre of the sphere



$N\cos\theta = mg$

($N\cos\theta$ is the component of total normal reaction force exerted in the vertical direction)

Friction force will act on the ball along the periphery of the hole

$$= \mu_k \times N = \mu_k \frac{mg}{\cos\theta}$$

It will act tangentially on the ball along the circle of radius r in the plane of the slab. Torque due to friction force about rotation axis $= \mu_k \frac{mg}{\cos\theta} \times r$

Angular impulse in time Δt

$$J = \tau_F \times \Delta t = \mu_k \frac{mg}{\cos\theta} \times r \times \Delta t$$

Ball comes to rest in the time Δt

$$\Rightarrow \quad J_{Friction} = \left|\Delta\vec{L}_{sphere}\right| \quad \Rightarrow \quad \mu_k \frac{mg}{\cos\theta} \times r \times \Delta t = \frac{2}{5}mR^2\omega$$

So, for two different values of r = r_1 and r_2, θ will be different and Δt will be different for the same change in angular momentum.

$$\mu_k \frac{mg}{\cos\theta_1} r_1 \Delta t_1 = \frac{2}{5}mR^2\omega$$

$$\mu_k \frac{mg}{\cos\theta_2} \times r_2 \Delta t_2 = \frac{2}{5}mR^2\omega$$

$$\frac{r_1\Delta t_1}{\cos\theta_1} = \frac{r_2\Delta t_2}{\cos\theta_2} \Rightarrow \quad \Delta t_2 = \frac{r_1\Delta t_1}{r_2}\frac{\cos\theta_2}{\cos\theta_1} = \frac{r_1\Delta t_1}{r_2}\frac{\sqrt{R^2 - r_2^2}}{\sqrt{R^2 - r_1^2}}$$

$$\Delta t_2 = \frac{8}{6} \times 9 \times \frac{\sqrt{1-(6/10)^2}}{\sqrt{1-(8/10)^2}} = 12 \times \left(\frac{64}{36}\right)^{\frac{1}{2}} = 12 \times \frac{8}{6} = 16\,\text{s}$$

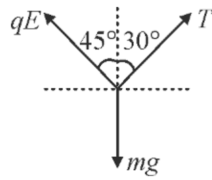
17.(5) $E_n - E_1 = \Delta E_1 + \Delta E_2$

$\therefore \quad \left[-\frac{13.6}{n^2}+13.6\right](z)^2 = \frac{12375}{1085}+\frac{12375}{304}$ Putting $Z=2$ for He^+, we get $n=5$

18.(0.02) $T\cos 30^\circ + qE\cos 45^\circ = mg$

$T\sin 30^\circ = qE\sin 45^\circ$

Solve to get $q = 20\times 10^{-3}C$



19.(6) $\bar{F} = 2xy^2\hat{i} + y\hat{j}$

$d\bar{r} = dx\hat{i} + dy\hat{j}$

$dw = \bar{F}.d\bar{r} = 2xy^2dx + ydy \quad = 2x.x^4dx + ydy$

$$w = \int_0^1 2x^5dx + \int_0^1 ydy = \frac{2}{6}\left[x^6\right]_0^1 + \left[\frac{y^2}{2}\right]_0^1 = \frac{2}{6}+\frac{1}{2} = \frac{5}{6}$$

20.(2) For small value of ω, friction will be directed radially inward as the tension in the string is zero. The string will develop tension only if the centripetal force required exceeds the limiting friction f_s. When block is stationary relative to platform,

Then, $T\sin 45^\circ = f\sin\theta \qquad \ldots$(i)

$T\cos 45^\circ + f\cos\theta = m\omega^2 r \qquad \ldots$(ii)

and for $\omega = \omega_{\max}$

$f = f_l = \mu mg$

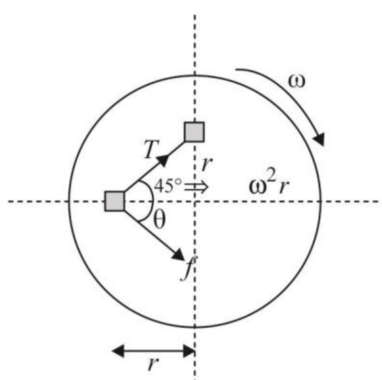
From (i) and (ii)

$f\sin\theta + f\cos\theta = m\omega^2 r$

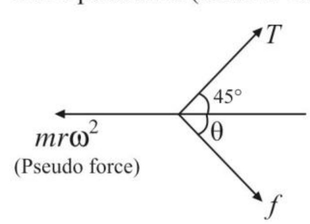
$\mu mg(\sin\theta + \cos\theta) = m\omega^2 r$

$$\omega^2 = \frac{(\sin\theta+\cos\theta)\mu g}{r}$$

$$\omega^2 = \sqrt{2}\sin(\theta+45^\circ)\,\frac{\mu g}{r}$$



For equilibrium (relative to platform)



For $\omega = \omega_{\max}$, $\sin(\theta+45^\circ) = 1$

$$\therefore \quad \omega_{\max}^2 = \sqrt{2}\,\frac{\mu g}{r} = \sqrt{2}\times\frac{1}{\sqrt{2}}\times\frac{10}{2.5} = 4$$

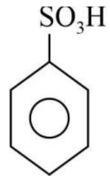
$\therefore \quad \omega_{\max} = 2$

CHEMISTRY

1.(D) $CH_3OH+\frac{3}{2}O_2 \rightarrow CO_2+2H_2O$

$\Delta G=-394.4+2(-237.1)+166.3=-702.3\,kJ$

2.(D) The Rate of SN1 reaction depends upon the stability of carbocation.

3.(A) 

, Rest all have pKa value greater than H_2CO_3

4.(D) $P=[CoCl_4]^{2-} \rightarrow$ unpaired electron is 3.

$Q=[Fe(CN)_6]^{4-} \rightarrow$ unpaired electron is 0.

$R=[MnCl_4]^{2-} \rightarrow$ unpaired electron is 5.

5.(B) Order w.r.t A = 1

Order w.r.t B = 2

Overall order = 3

6.(C) $S_2O_3^{2-} \;+\; 2H^+ \xrightarrow{\Delta} \underset{\text{Pungent odour}}{SO_2(g)} \;+\; \underset{\text{pale yellow ppt.}}{S\downarrow} \;+ H_2O$

7.(A) $\frac{r_{A^+}}{r_{B^-}} \geq 0.225$

$r_{A^+} \geq 0.225\times180$

$r_{A^+} \geq 40.5 \cong 41$

8.(A) $Cd^{2+}(aq)+2NH_3(aq)+2H_2O(l) \rightarrow Cd(OH)_2(s)+2NH_4^+(aq)$

$Cd(OH)_2(s)+4NH_3(aq) \rightarrow \underset{\text{(soluble)}}{[Cd(NH_3)_4]^{2+}_{(aq)}}+2OH^-(aq)$

9.(B) At 83K Nitrogen gas physically adsorbed. Raising temperature to 773K, then results in the nitrogen being chemisorbed.

10.(A) Common ore of $Mg-MgCO_3$

$Zn-ZnCO_3$

$Mn-MnCO_3$

$Fe-FeCO_3$

$Ag-Ag_2S$

$Al-Al_2O_3.2H_2O$

$Na-NaCl$

11.(AB) Solution is an example of positive deviation, so that ΔH_{mixing} is positive and ΔS_{system} is always positive for mixing of solution

12.(AB) (A) and (B). exhibit (cis-trans) isomerism

13.(B) $K_a\left(C_6H_5COOH\right)=1\times10^{-4}$

pH of 0.01M C_6H_5COONa

$$\underset{0.01(1-h)}{C_6H_5COO^-}+H_2O \rightleftharpoons \underset{0.01h}{C_6H_5\,COOH}+\underset{0.01h}{OH^-}$$

$$k_h=\frac{k_w}{k_a}=\frac{0.01h^2}{1-h}$$

$$\frac{10^{-14}}{10^{-4}}=\frac{10^{-2}h^2}{1-h}\quad(1-h\simeq1)$$

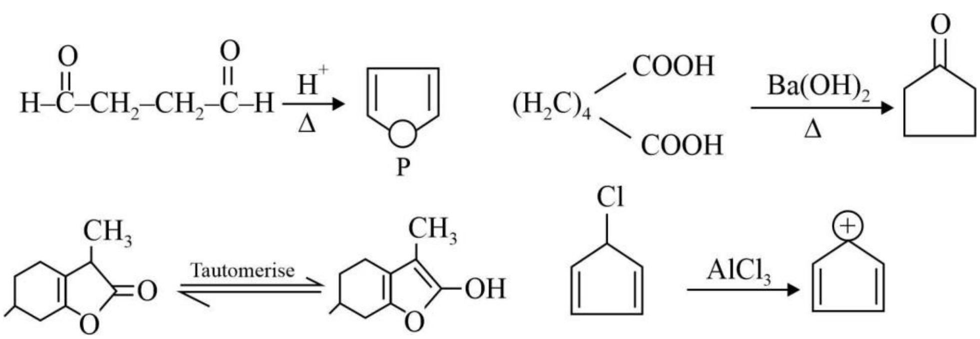
$$\left[OH^-\right]=0.01h=0.01\times10^{-4}=10^{-6}$$

$$\left[H^+\right]=10^{-8}$$

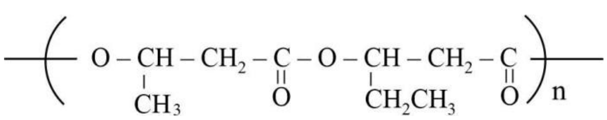
pH = 8.

14.(AB) In hyperconjugation σ electron of hydrogen atom interact with vacant p-orbital of carbon atom

15.(AC)



16.(8) Structure of PHBV polymer



17.(6) The central chromium atom is in +6 state in the given complex.

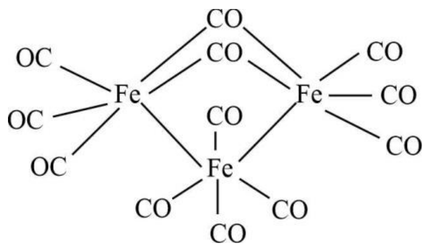
18.(4) $\lambda=\dfrac{h}{\sqrt{2m(KE)}}$ $\qquad$ KE $\propto$ T

$$\frac{\lambda_{He}}{\lambda_{proton}}=\sqrt{\frac{m_{proton}KE_{proton}}{m_{He}KE_{He}}}=\sqrt{\frac{1\times50}{4\times200}}=\frac{1}{4}$$

$$M=\frac{1}{4}$$

$$\text{Hence}=\frac{1}{M}=4$$

19.(2)



20.(4) Leucine form N-terminal then possible combination

Leu — Ala–phen – gly

Leu — phen – Ala – gly

Leu — Ala – Gly – phe

Leu — phen – Gly – Ala

MATHEMATICS

1.(B) We have, $\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right)+\tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right)+\tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right)+...+\tan^{-1}\left(\frac{1}{c_n}\right)$

$$=\tan^{-1}\left(\frac{\frac{x}{y}-\frac{1}{c_1}}{1+\frac{x}{y}\frac{1}{c_1}}\right)+\tan^{-1}\left(\frac{\frac{1}{c_1}-\frac{1}{c_2}}{1+\frac{1}{c_1c_2}}\right)+\tan^{-1}\left(\frac{\frac{1}{c_2}-\frac{1}{c_3}}{1+\frac{1}{c_2c_3}}\right)+....+\tan^{-1}\left(\frac{1}{c_n}\right)$$

$$=\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\left(\frac{1}{c_1}\right)+\tan^{-1}\left(\frac{1}{c_1}\right)-\tan^{-1}\left(\frac{1}{c_2}\right)+\tan^{-1}\left(\frac{1}{c_2}\right)-\tan^{-1}\left(\frac{1}{c_3}\right)$$

$$+...-\tan^{-1}\left(\frac{1}{c_n}\right)+\tan^{-1}\left(\frac{1}{c_n}\right)=\tan^{-1}\left(\frac{x}{y}\right)$$

2.(B) Let length of the edge of the regular tetrahedron be L. Let O be (0, 0, 0)

$A : L\hat{i}$

$B : L\left(\cos\left(\frac{\pi}{3}\right)\hat{i}+\sin\left(\frac{\pi}{3}\right)\hat{j}\right)$

$C : L\left(\frac{1}{2}\hat{i}+\frac{1}{2\sqrt{3}}\hat{j}+\sqrt{\frac{2}{3}}\hat{k}\right)$

It follows that

$P : L\left(\frac{1}{3}\hat{i}\right)$

$Q : \frac{L}{2}\left(\cos\left(\frac{\pi}{3}\right)\hat{i}+\sin\left(\frac{\pi}{3}\right)\hat{j}\right)$

$R : \frac{L}{3}\left(\frac{1}{2}\hat{i}+\frac{1}{2\sqrt{3}}\hat{j}+\sqrt{\frac{2}{3}}\hat{k}\right)$

$Ar\,(\Delta OAB)=\frac{\sqrt{3}}{4}L^2$

$Ar\,(\Delta PQR)=\frac{1}{2}|\overrightarrow{PQ}\times\overrightarrow{PR}|=\frac{L^2}{6\sqrt{6}}$

3.(C) $\because$ Centre of the circle $|z-2|=2$ *ie*, 2

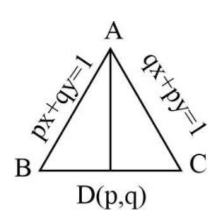
lie on $z(1-i)+\bar{z}(1+i)=4,$

Hence given line $z(1-i)+\bar{z}(1+i)=4$ pass through the centre of circle i. e, intersect at two points.

$\therefore$ Number of solutions = 2

4.(C) Equation of line through AB and AC is $(px+qy-1)+\lambda(qx+py-1)=0$ …(1)

$\because$ It is passes through (p, q), then $(p^2+q^2-1)+\lambda(pq+pq-1)=0$



$\therefore$ $\lambda=-\frac{(p^2+q^2-1)}{(2pq-1)}$

From Eq. (i),

$$(px+qy-1)-\frac{(p^2+q^2-1)}{(2pq-1)}(qx+py-1)=0$$

or $(2pq-1)(px+qy-1)=(p^2+q^2-1)(qx+py-1)$

5.(B) Coordinates of any point Q on the given line are $(2r+1,-3r-1,8r-10)$.

So the direction cosines of PQ *are* $2r,-3r-1,8r-10$.

Now PQ is perpendicular to the given line.

if $\quad 2(2r)-3(-3r-1)+8(8r-10)=0$

$\Rightarrow \quad 77r-77=0 \Rightarrow r=1$

and the coordinates of Q, the foot of the perpendicular from P on the line are $(3,-4,-2)$.

Let $R\ (a,b,c)$ be the reflection of P in the given lines then Q is the mid-point of PR

$\Rightarrow \quad \frac{a+1}{2}=3, \frac{b}{2}=-4, \frac{c}{2}=-2$

$\Rightarrow \quad a=5,\ b=-8,\ c=-4$

6.(B) Out of the numbers 10, 11, 12, 13, …., 99 those numbers the product of whose digits is 12 are 26, 34, 43, 62 ie, only 4.

$\therefore \quad p=p(E)=\frac{4}{90}=\frac{2}{45}$

$q=P(\overline{E})=1-P(E)=1-\frac{2}{45}=\frac{43}{45}$

Hence, the probability that he will laugh atleast once = $1-q^3$

$=1-\left(\frac{43}{45}\right)^3$

7.(D) $\because\ f(x)=\max\left\{\sin x, \cos x, \frac{1}{2}\right\}$

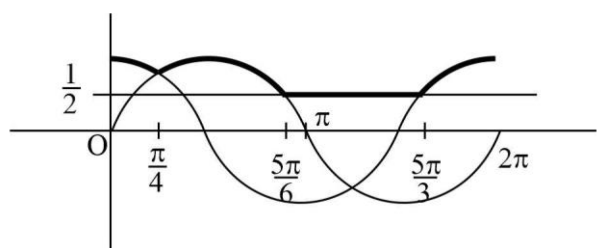
Interval value of $f(x)$

For $\quad 0\le x<\frac{\pi}{4}, \cos x$

For $\quad \frac{\pi}{4}\le x<\frac{5\pi}{6}, \sin x$

For $\quad \frac{5\pi}{6}\le x<\frac{5\pi}{3}, \frac{1}{2}$

For $\quad \frac{5\pi}{3}\le x<2\pi, \cos x$



Hence, required area = $\int_0^{\pi/4}\cos x dx+\int_{\pi/4}^{5\pi/6}\sin x dx+\int_{5\pi/6}^{5\pi/3}(1/2)dx+\int_{5\pi/3}^{2\pi}\cos x dx$

$=[\sin x]_0^{\pi/4}-[\cos x]_{\pi/4}^{5\pi/6}+\frac{1}{2}[x]_{5\pi/6}^{5\pi/3}+[\sin x]_{5\pi/3}^{2\pi}$

$=\left(\frac{1}{\sqrt{2}}-0\right)-\left(-\frac{\sqrt{3}}{2}-\frac{1}{\sqrt{2}}\right)+\frac{1}{2}\left(\frac{5\pi}{3}-\frac{5\pi}{6}\right)+\left(0+\frac{\sqrt{3}}{2}\right)=\left(\frac{5\pi}{12}+\sqrt{2}+\sqrt{3}\right)$ *sq unit.*

8.(B) The given equation can be written as $(x\,dy + y\,dx) + \left\{\ln x\,dy + \frac{y}{x}dx\right\} + (\sin y\,dx + x\cos y\,dy) = 0$

$\Rightarrow \quad d(xy) + d(y\ln x) + d(x\sin y) = 0$

On integrating, we get $xy + y\ln x + x\sin y = c$

Where c is arbitrary constant

9.(D) $\because \; f(x) = \int_2^{x^2} \frac{\left(\sin^{-1}\sqrt{t}\right)^2}{\sqrt{t}}\,dt$

$\therefore \; f'(x) = \frac{\left(\sin^{-1}x\right)^2}{x}(2x - 0)$

$f'(x) = 2\left(\sin^{-1}x\right)^2 \qquad \ldots(i)$

$f''(x) = \frac{4\sin^{-1}x}{\sqrt{(1-x^2)}}$

$\therefore \; [f''(x)]^2 = \frac{16\left(\sin^{-1}x\right)^2}{(1-x^2)} \qquad \ldots(ii)$

From Eqs. (i) and (ii), we have $(1-x^2)[f''(x)]^2 - 2f'(x) = 12\left(\sin^{-1}x\right)^2 \; = 12\left(\sin^{-1}\frac{1}{\sqrt{2}}\right)^2 \; = \frac{3}{4}\pi^2$

10.(B) $\sin^7 y = \left|x^3 - x^2 - 9x + 9\right| + \left|x^3 - x^2 - 4x + 4\right| + \sec^2 2y + \cos^4 y$

For $\quad x = 1$

$\sin^7 y = \sec^2 2y + \cos^4 y$

$\Rightarrow \quad \sin^7 y\cos^2 2y = 1 + \cos^4 y\cos^2 2y$

Since, LHS ≤ 1.

Which is possible only when

$\therefore \quad \sin^7 y\cos^2 2y = 1$

$\Rightarrow \quad \sin^7 y = 1 \; and \; \cos^2 2y = 1$

$y = \frac{\pi}{2}$

General value of y is $2n\pi + \frac{\pi}{2}$

Hence, $x = 1$ and $y = 2n\pi + \frac{\pi}{2}, n \in I$

11.(ABC) The function *f* is not differentiable at $x = 0$,

$x = \pi/2 \; as \; f'(0-) = -10, f'(0+) = 1; f'(\pi/2-) = 0, f'(\pi/2+) = -1.$

The function $f'(x)$ is given by

$$f'(x) = \begin{cases} 3x^2 + 2x - 10 & -1 \le x < 0 \\ \cos x & 0 < x < \pi/2 \\ -\sin x & \pi/2 < x \le \pi \end{cases}$$

The critical points of f are given by $f'(x)=0$ *or* $x=0, \pi/2$. Thus critical points are $x=\pi/2, x=0\ x=\pi$. Since $f'(x)>0,$ for $0<x<\pi/2$ and $f'(x)<0,$ for $\pi/2<x<\pi$ so f has local maxima at $x=\pi/2$. Also $f'(x)<0$ for $-1\le x<0$ and $f'(x)>0$ for $0<x<\pi/2$ so f has local minima at $x=0$. Since $f(-1)=10,\ f(\pi/2)=1,\ f(0)=0$ and $f(\pi)=0$. Thus f has absolute maximum at $x=-1$ and absolute minimum at $x=0,\ \pi$.

12.(ABC)

$$\frac{1}{n+1}+\frac{1}{n+2}+...+\frac{1}{2n}=\left(1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{2n}\right)-\left(1+\frac{1}{2}+....+\frac{1}{n}\right)$$

$$=\left(1+\frac{1}{3}+....+\frac{1}{2n-1}\right)+\frac{1}{2}\left(1+\frac{1}{2}+....+\frac{1}{n}\right)-\left(1+\frac{1}{2}+...+\frac{1}{n}\right)$$

$$=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}-....+\frac{1}{2n-1}-\frac{1}{2n}$$

Also, $\quad a(2n)<\underbrace{\frac{1}{n}+\frac{1}{n}+...+\frac{1}{n}}_{n\text{ times}}=1$

and $\alpha(2n)\ge\frac{1}{2n}\ge\frac{1}{2}\forall n.$

13.(ABC) Equation $r(t)=R(u)$ we obtain $(1+t-2u)\hat{i}+(-10+2t-u)\hat{j}+(1+t-2u)\hat{k}=0$

Thus $\quad 1+t-2u=0;-10+2t-u=0$

Solving, we obtain $t=7, u=4,$ *so* $\vec{r}(7)=8\hat{i}+8\hat{j}+9\hat{k}=\vec{R}(4)$

The two lines intersect at the tip of this vector

14.(BD) $(1-\tan\theta)(1+\tan\theta)\sec^2\theta+2^{\tan^2\theta}=0$

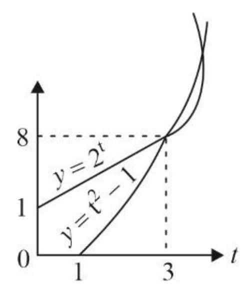
$\Rightarrow \quad (1-\tan^4\theta)+2^{\tan^2\theta}=0$

$\Rightarrow \quad 2^t=t^2-1$ (where $t=\tan^2\theta$)

By inspection (or by graph) we find $y=2^t$ *and* $y=t^2-1$ intersect in $t=3$.

$\Rightarrow \quad \tan^2\theta=3\Rightarrow\tan\theta=\pm\sqrt{3}$

$\Rightarrow \quad \theta=\pm\pi/3\ \&\ \theta=\pm\alpha,\ \alpha\in\left(\frac{\pi}{3},\frac{\pi}{2}\right)$



15.(AC) $AB=\frac{1}{\left|(AB)^{-1}\right|}adj((AB)^{-1})$

$=|AB|\,adj(B^{-1}A^{-1})$

$=|A||B|\,adj(A^{-1})\,adj(B^{-1})$

16.(5) Let the equation of the ellipse is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (let $a>b$)

Then the foci are $S(ae,0)$ and $S'(-ae,0)$ length of minor axis $BB'=2b$

and length of latusrectum = $\frac{2b^2}{a}$

$\therefore \quad$ According to the question BB´ = SS´

$\Rightarrow \quad 2b=2ae$

$\Rightarrow \quad b = ae \quad \ldots(i)$

and $\quad \dfrac{2b^2}{a} = 10$

$\Rightarrow \quad b^2 = 5a \quad \ldots(ii)$

Also, we have $b^2 = a^2\left(1-e^2\right)$ Putting the value of b from Eq. (i)

In Eq. (iii), we have $\quad a^2e^2 = a^2\left(1-e^2\right)$

$\Rightarrow \quad e^2 = 1-e^2$

$\Rightarrow \quad 2e^2 = 1$

$\therefore \quad e = \dfrac{1}{\sqrt{2}}$

From Eq. (i), we have $b = \dfrac{a}{\sqrt{2}}$

$\therefore \quad b^2 = \dfrac{a^2}{2}$

$\Rightarrow \quad 5a = \dfrac{a^2}{2}$ [from Eq. (ii)]

$\Rightarrow \quad a = 10$

From Eq. (ii) $\quad b^2 = 5\times 10 = 50$

Putting the values of a and b in $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1,$ the equation of required ellipse is $\dfrac{x^2}{100}+\dfrac{y^2}{50}=1$

$x^2+2y^2=100$

$\therefore \quad m=1, n=2$

Then, $\quad 10^m+20^n=10+400=410$

17.(3.09)

$\because \quad a_r = r^2.\dfrac{(100-r+1)}{r} = r.(101-r)$

$\because \quad \prod_{i=1}^{100}(x-a_i) = (x-a_1)(x-a_2)(x-a_3)...(x-a_{100})$

$= x^{100}-(a_1+a_2+a_3+...+a_{100})x^{99}...$

$\therefore \quad \lambda = -(a_1+a_3+...+a_{100}) = -\sum_{r=1}^{100}a_r = -\sum_{r=1}^{100}r.(100-r) = -100\sum_{r=1}^{100}r+\sum_{r=1}^{100}r^2$

18.(1) $\because \quad \overrightarrow{AB} = \left(2\hat{i}+3j-4k\right)-\left(3\hat{i}-2j-k\right) = -\hat{i}+5j-3k$

$\overrightarrow{AC} = \left(-\hat{i}+j+2k\right)-\left(3\hat{i}-2j-k\right) = -4\hat{i}+3j+3k$

and $\overrightarrow{AD} = \left(4\hat{i}+5j+\lambda k\right)-\left(3\hat{i}-2j-k\right) = \hat{i}+7j+(\lambda+1)k$

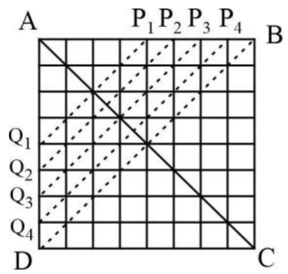
Vectors $\overrightarrow{AB}, \overrightarrow{AC}$ *and* $\overrightarrow{AD}$ will be coplanar

$\Rightarrow \quad \begin{vmatrix} -1 & 5 & -3 \\ -4 & 3 & 3 \\ 1 & 7 & \lambda+1 \end{vmatrix} = 0 \quad \Rightarrow \quad 17\lambda+146=0 \quad \therefore \quad -17\lambda = 146$

19.(2) Total number of ways = ${}^{64}C_4$

The chess board can be divided into two parts by a diagonal line BD. Now, if we begin to select four squares from the diagonal $P_1Q_1, P_2Q_2, ..., BD,$ then we can find number of squares selected

$$= 2\left({}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4\right) + {}^8C_4 = 182$$



and similarly number of squares for the diagonals chosen parallel to AC = 182

$\therefore$ Total favourable ways = 364

$\therefore$ Required probability = $\dfrac{364}{{}^{64}C_4}$

$\therefore$ $\lambda = 364$

20.(5) Let p and p + 1 be removed number from 1, 2, …, n. Then arithmetic mean of the remaining numbers is

$$\frac{\frac{n(n+1)}{2} - (p + p + 1)}{n - 2} = \frac{105}{4} \qquad \text{(given)}$$

$\therefore$ $2n^2 - 103n - 8p + 206 = 0$

Since, n and p are integers, so n must be even let $n = 2r$

We get, $p = \dfrac{4r^2 + 103(1 - r)}{4} = r^2 + \dfrac{103(1 - r)}{4}$

Since, p is an integer, then (1 – r) must be divisible by 4.

Let $r = 1 + 4t,$ we get $n = 2 + 8t$

and $p = 16t^2 - 95t + 1,$ Now $1 \le p < n$

$\Rightarrow$ $1 \le 16t^2 - 95t + 1 < 8t + 2$

$\Rightarrow$ $t = 6$

$\therefore$ $n = 2 + 8 \times 6 = 50$